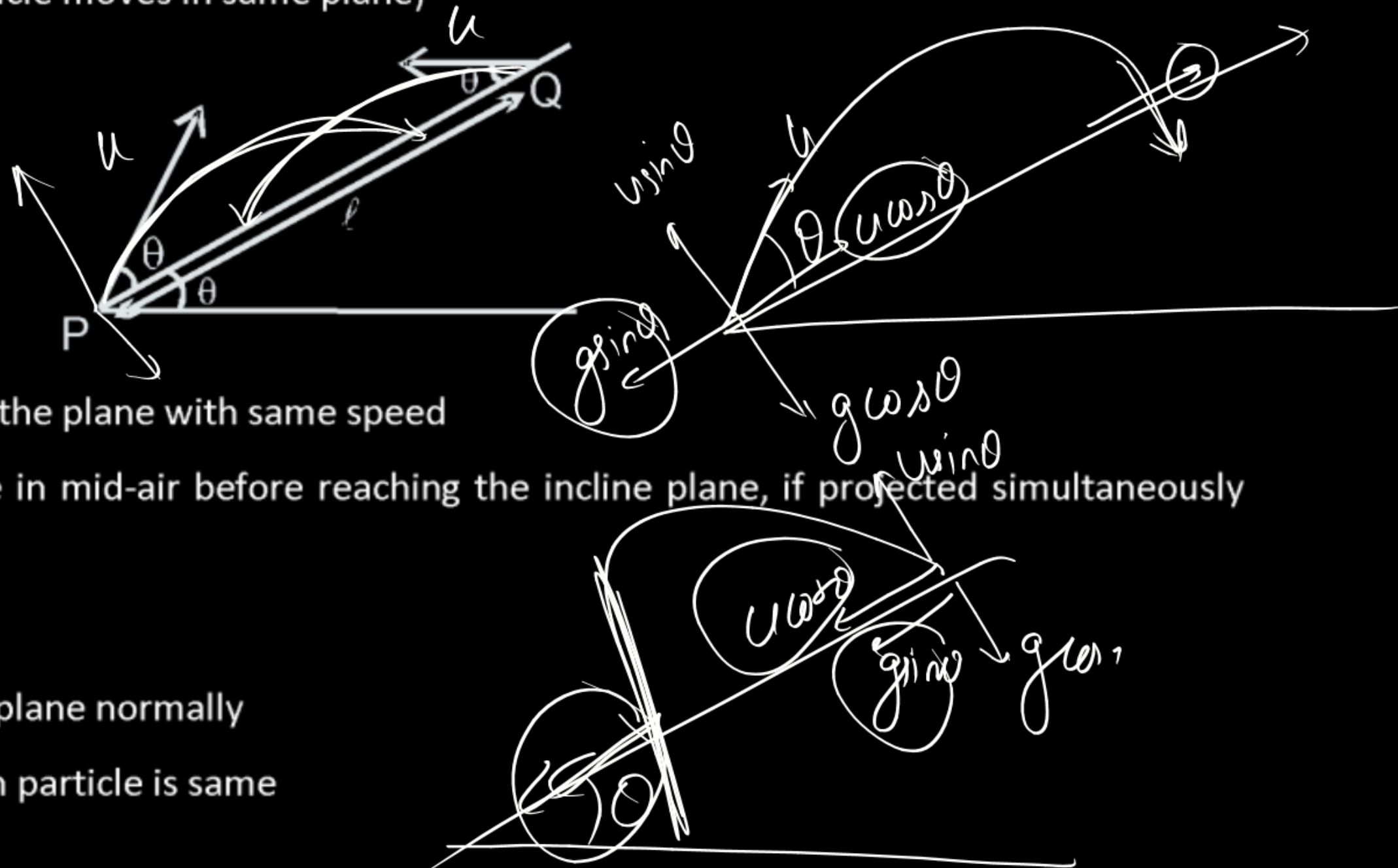


From an incline plane two particles P and Q are projected with same speed at same angle  $\theta$ , one up and other down the incline plane as shown in figure. Which of the following statement(s) is/are correct? (Assume both particle moves in same plane)

$$\frac{2u \sin \theta}{g}$$



☒ (A) The particles will collide the plane with same speed

☒ (B) The particles will collide in mid-air before reaching the incline plane, if projected simultaneously

and  $\ell < \frac{4u^2 \sin \theta}{g}$

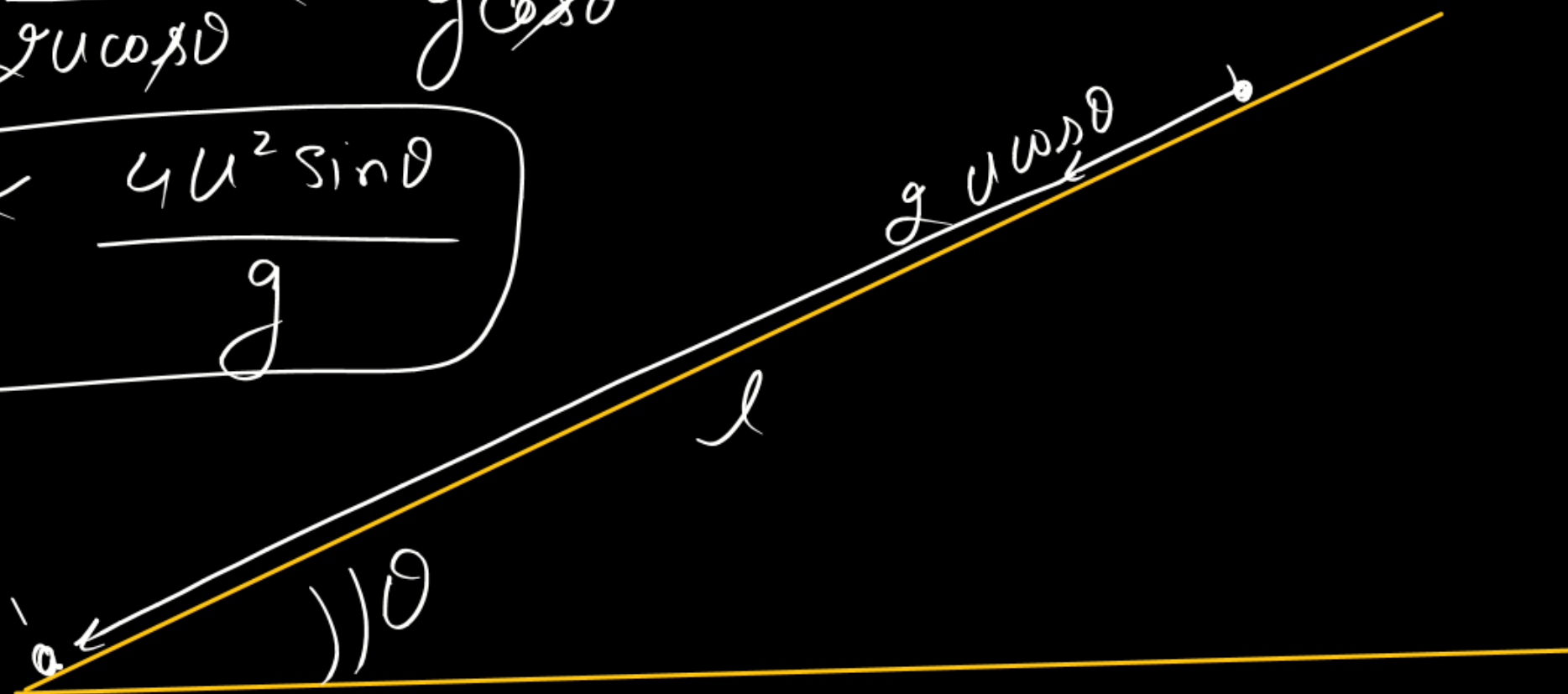
☒ (C) Both particles strike the plane normally

☒ (D) The time of flight of each particle is same

$$t = \frac{l}{g \cos \theta} < T$$

$$\frac{l}{g \cos \theta} < \frac{2u \sin \theta}{g \cos \theta}$$

$$l < \frac{4u^2 \sin \theta}{g}$$



A boat moves relative to river with a velocity which is  $n$  times the river flow velocity.

~~(A)~~ If  $n < 1$ , boat cannot cross the river

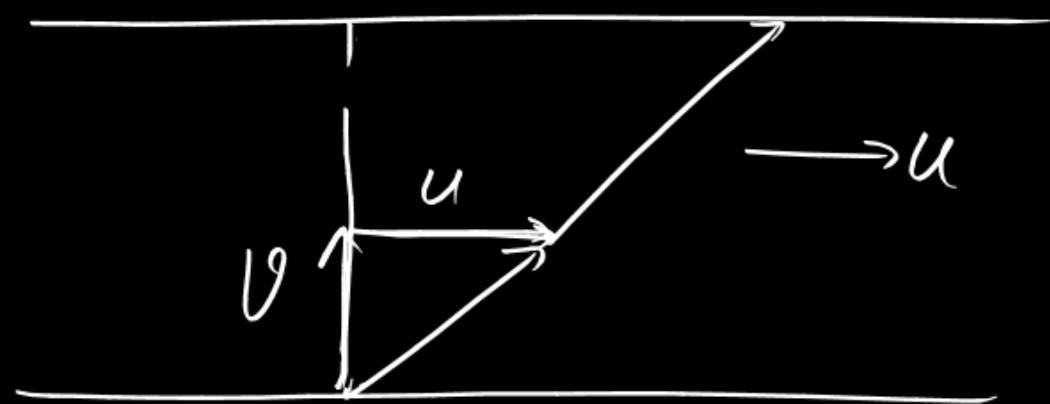
✓ (B) If  $n = 1$ , boat cannot cross the river without drifting

✓ (C) If  $n > 1$ , boat can cross the river along shortest path

(u)

$$v = nu$$

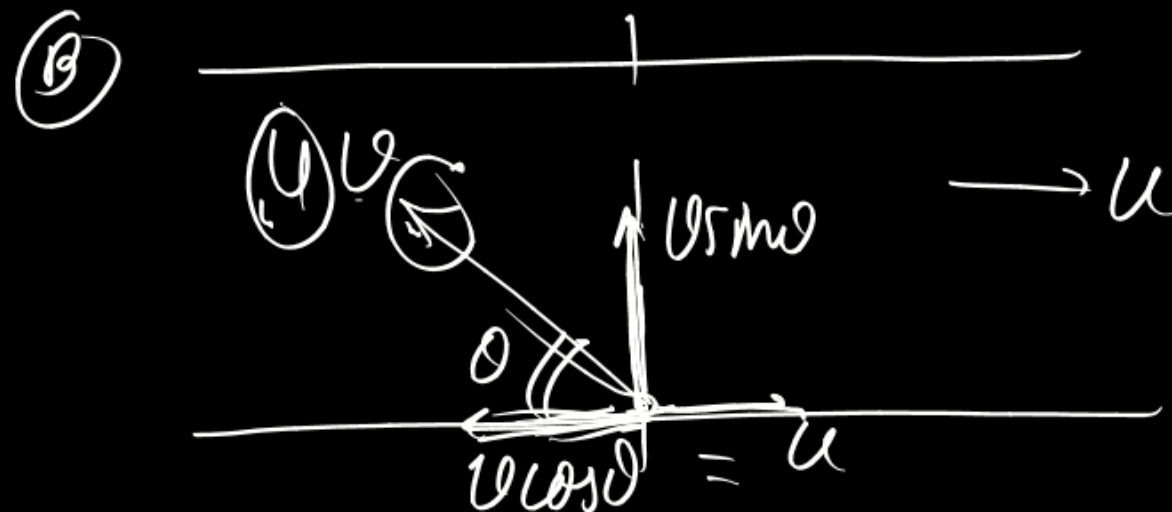
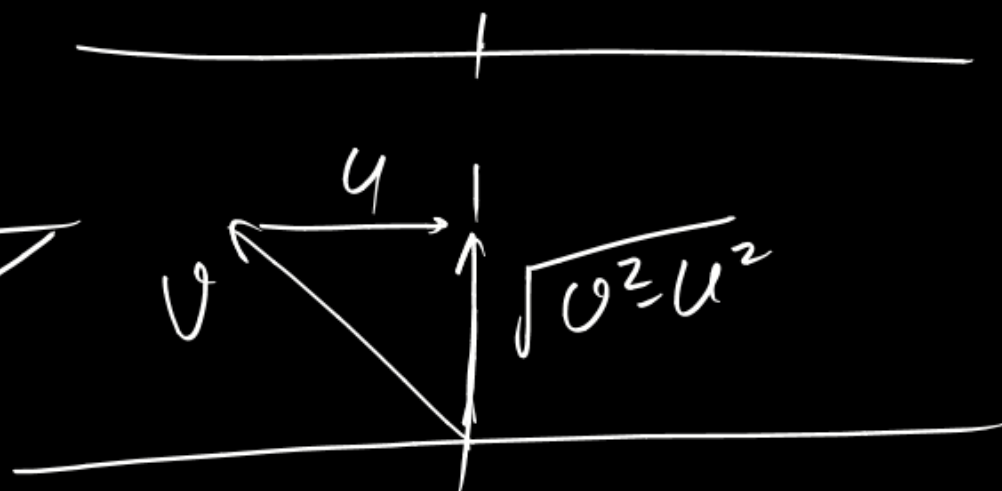
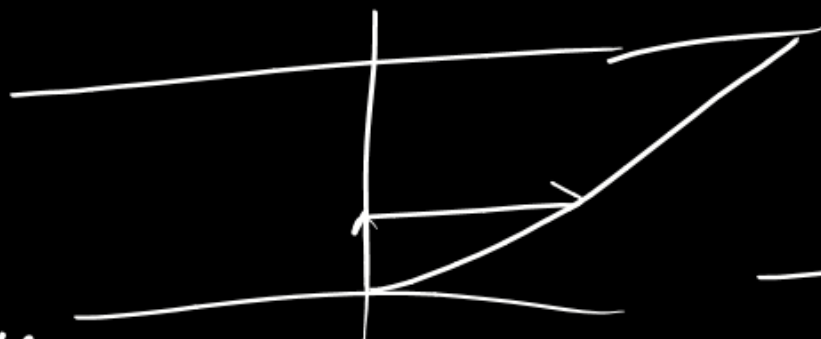
Ⓐ ✓ ~~(D)~~ Boat can cross the river whatever is the value of  $n$  (excluding zero)



$$n < 1$$

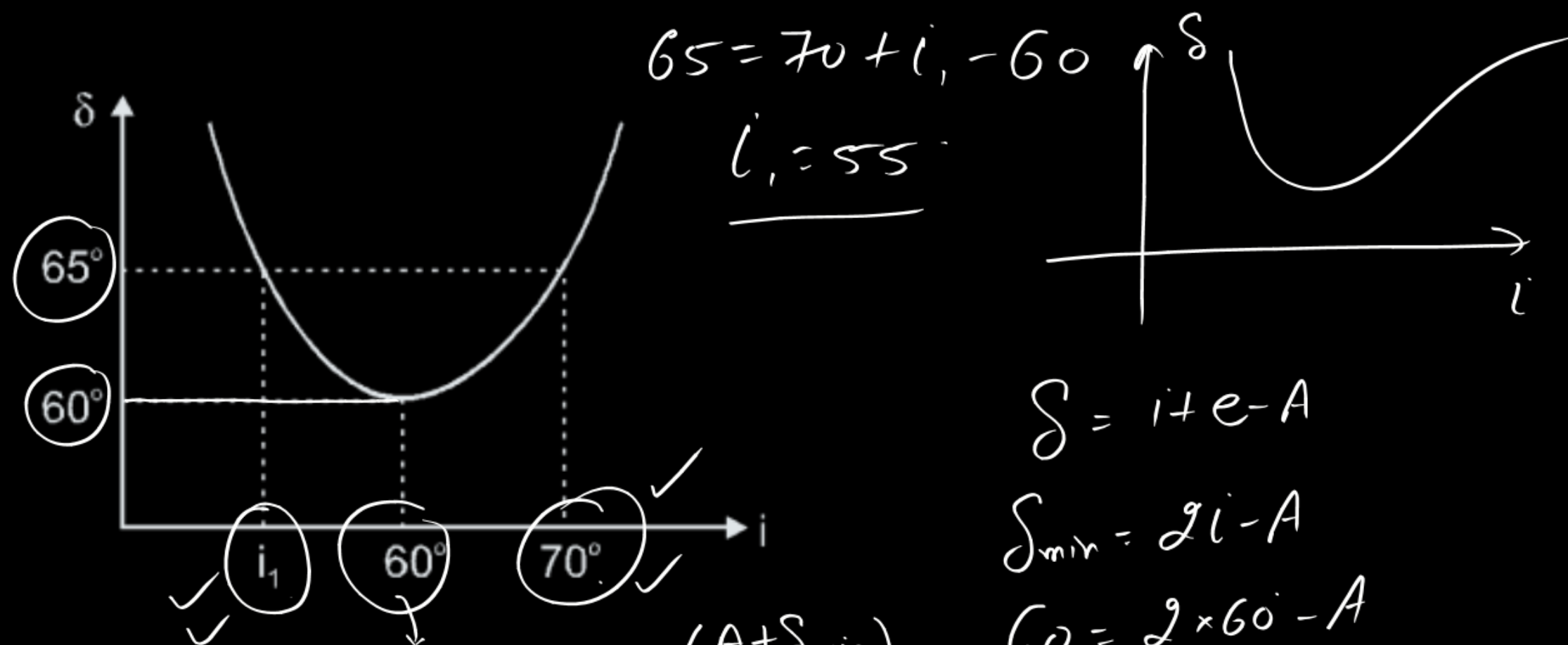
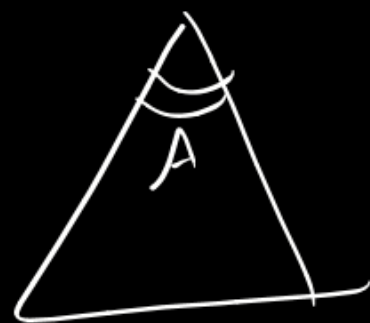
$$\underline{v < u}$$

Ⓒ  $v > u$



$$\underline{v = u}$$

The angle of deviation ( $\delta$ ) vs angle of incidence ( $i$ ) is plotted for a prism. Pick up the correct statements.



(A) The angle of prism is  $60^\circ$

(B) The refractive index of the prism is  $n = \sqrt{3}$

(C) For deviation to be  $65^\circ$  the angle of incidence  $i_1 = 55^\circ$

(D) The curve of ' $\delta$ ' vs ' $i$ ' is parabolic

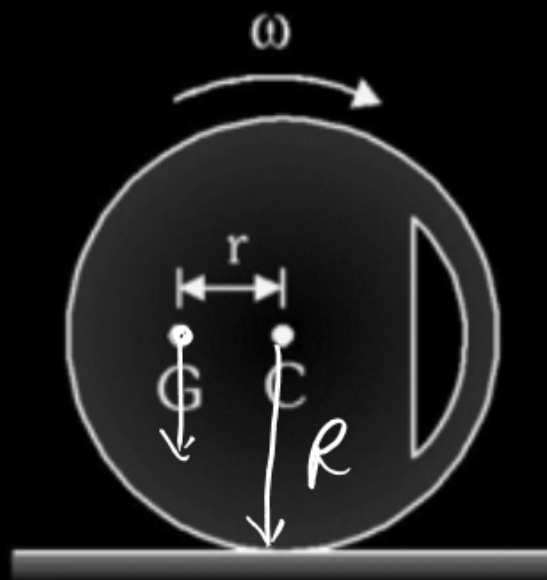
$$n = \frac{\sin\left(\frac{A + \delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$n = \frac{\sin 60}{\sin 30} = \frac{\sqrt{3} \times 2}{2 \times 1} = \sqrt{3}$$

The mass center G of a 3 kg wheel of radius  $R = 180$  mm is located at a distance  $r = 60$  mm from its geometric center C. The centroidal radius of gyration of the wheel is  $k = 90$  mm. As the wheel rolls without sliding, its angular velocity is observed to vary. Angular speed of wheel is  $\omega = 8$  rad/s in the position shown

$$g = 9.8 \text{ m/s}^2$$

$$I = mk^2$$



The angular velocity of the wheel when the mass center G is directly above the geometric center C is close to :

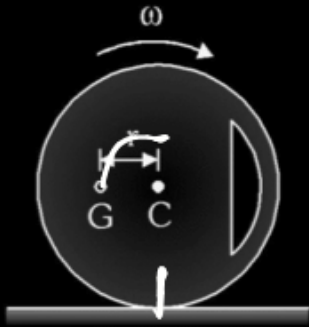
(A) 20 rad/s.

(B) 15 rad/s.

(C) 10 rad/s.

✓ (D) 5 rad/s.

The mass center G of a 3 kg wheel of radius  $R = 180$  mm is located at a distance  $r = 60$  mm from its geometric center C. The centroidal radius of gyration of the wheel is  $k = 90$  mm. As the wheel rolls without sliding, its angular velocity is observed to vary. Angular speed of wheel is  $\omega = 8$  rad/s in the position shown



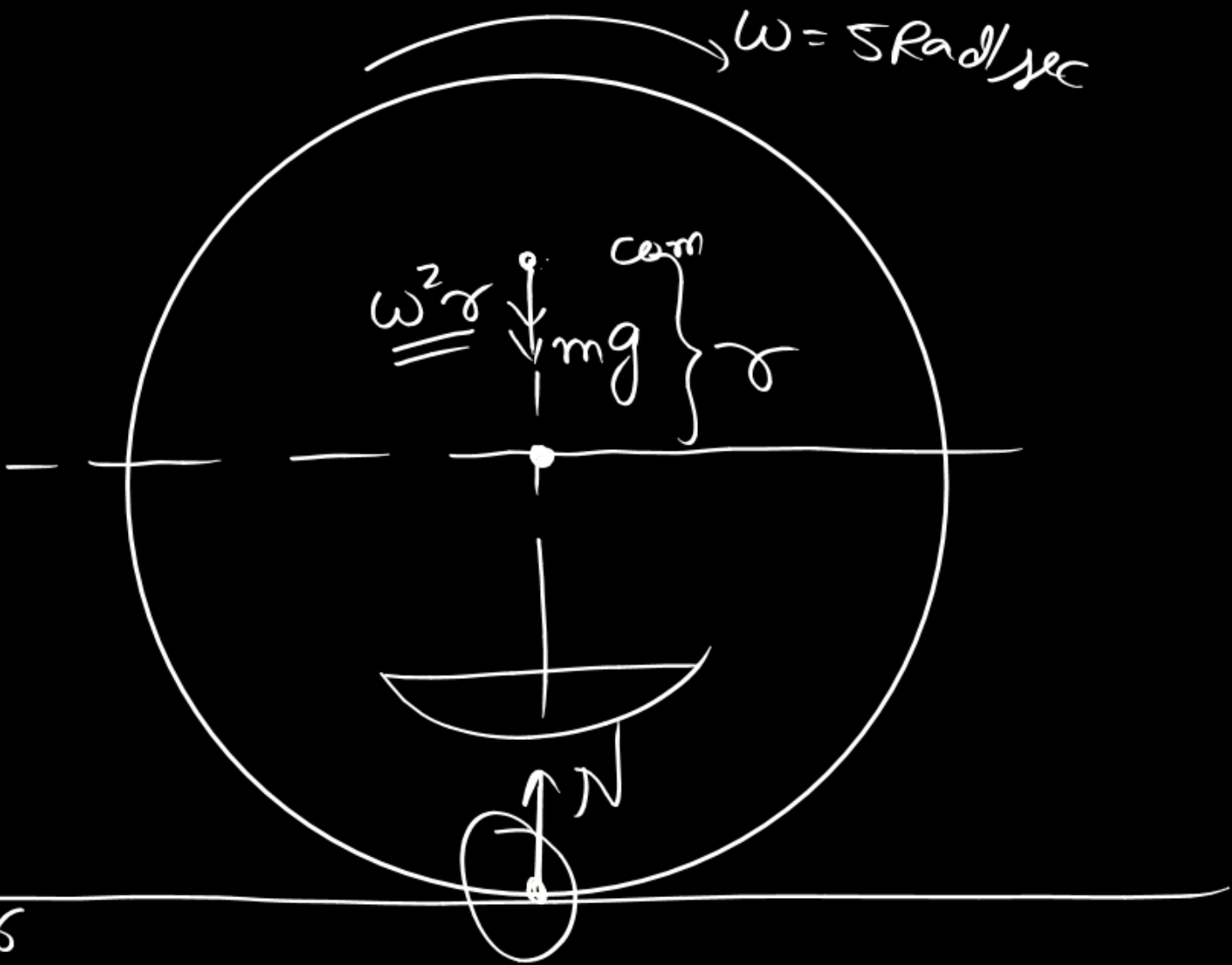
The reaction at the horizontal surface when the mass center G is directly above the geometric center C is close to:

- (A) 15 N
- (B) 60 N
- (C) 25 N
- (D) 40 N

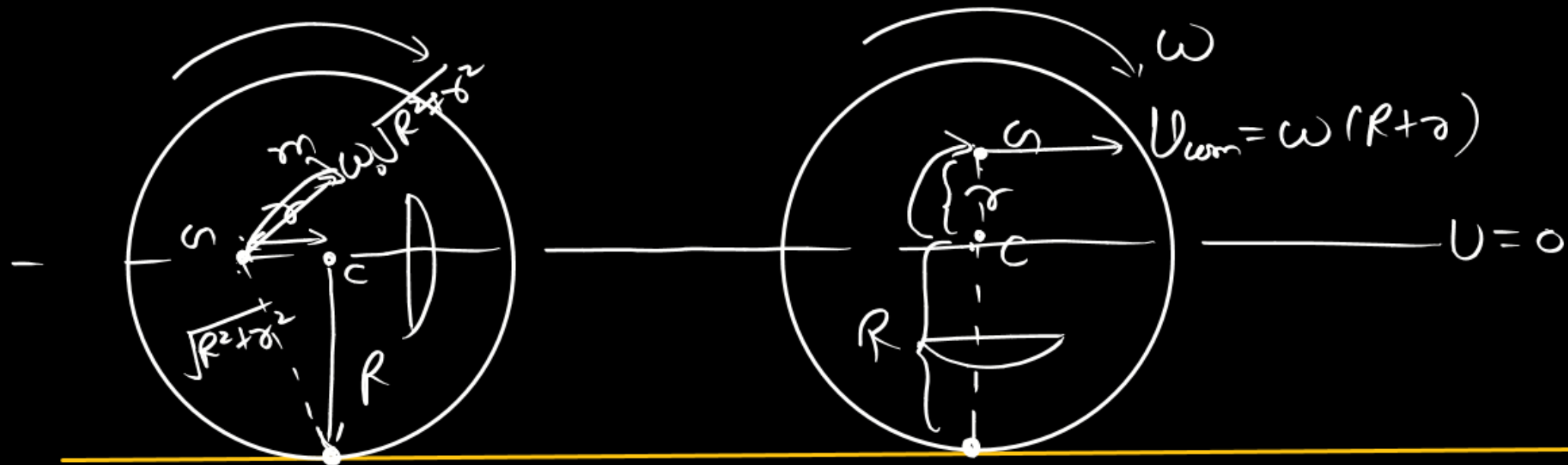
$$mg - N = m\omega^2 r$$

$$N = mg - m\omega^2 r$$

$$= 3 \times 9.8 - 3 \times 25 \times 0.06$$



$$\omega_0 = 8 \text{ rad/sec}$$



TMEC

$$K.E. = \frac{1}{2} m U_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} m \omega_0^2 (R^2 + r^2) + \frac{1}{2} m k^2 \omega_0^2$$

$$U = 0$$

$$K_i + U_i = K_f + U_f$$

$$\omega = 5 \text{ Rad/sec}$$

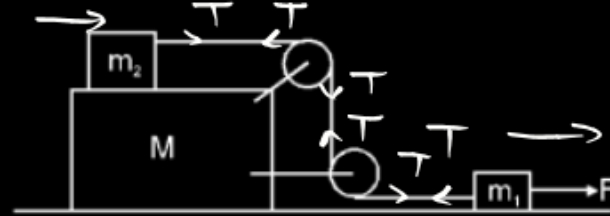
Final

$$K.E = \frac{1}{2} m \omega^2 (R+r)^2 + \frac{1}{2} m k^2 \omega^2$$

$$U = mg r$$

Match the following :

Three blocks of masses  $m_1$ ,  $m_2$  and  $M$  are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. Pulleys are light. A constant force  $F$  is applied on block of mass  $m_1$ . Pulleys and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with masses  $m_1$  and  $m_2$  are horizontal.



List-I

(P) Acceleration of mass  $m_1$

(Q) Acceleration of mass  $m_2$

(R) Acceleration of mass  $M$

(S) Tension in the string

List-II

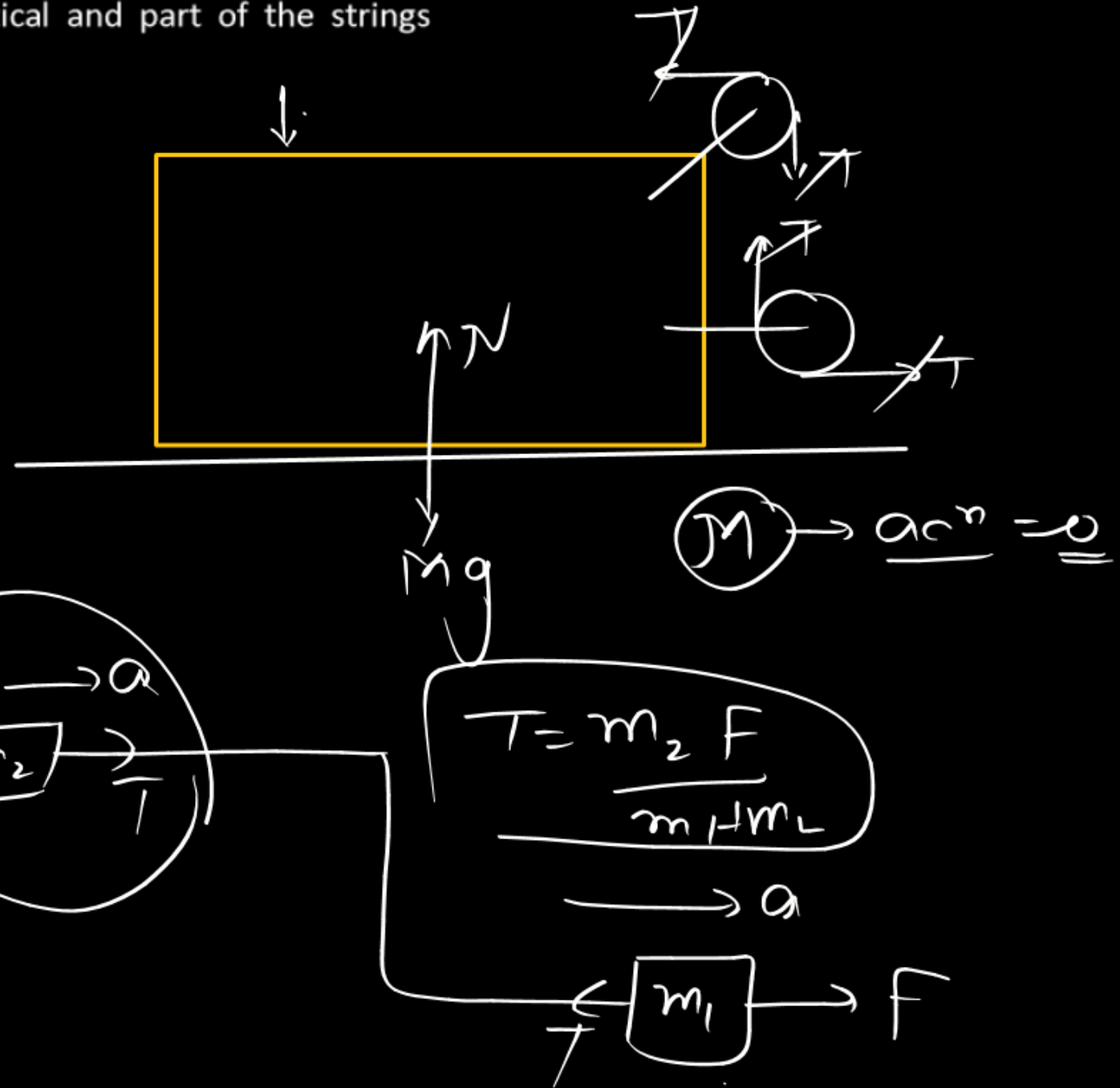
(1)  $\frac{F}{m_1}$

(2)  $\frac{F}{m_1 + m_2}$

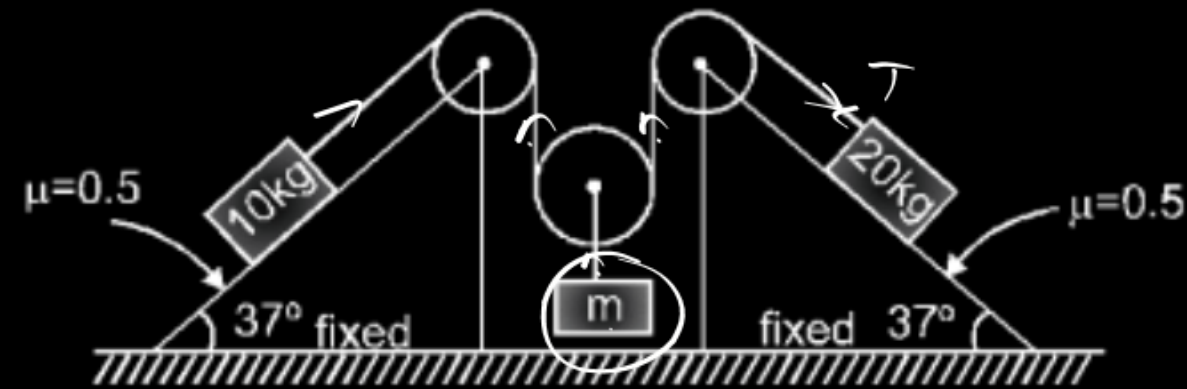
(3) zero

(4)  $\frac{m_2 F}{m_1 + m_2}$

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 2 | 3 | 4 |
| (B) | 2 | 1 | 3 | 4 |
| (C) | 2 | 2 | 4 | 1 |
| (D) | 2 | 1 | 3 | 1 |



In given arrangement, 10 kg and 20 kg blocks are kept at rest on two fixed inclined planes. All strings and pulleys are ideal. The difference of maximum and minimum values of  $m$  for which system remain in equilibrium is: ( $g = 10 \text{ m/s}^2$ )



$$2T = mg$$

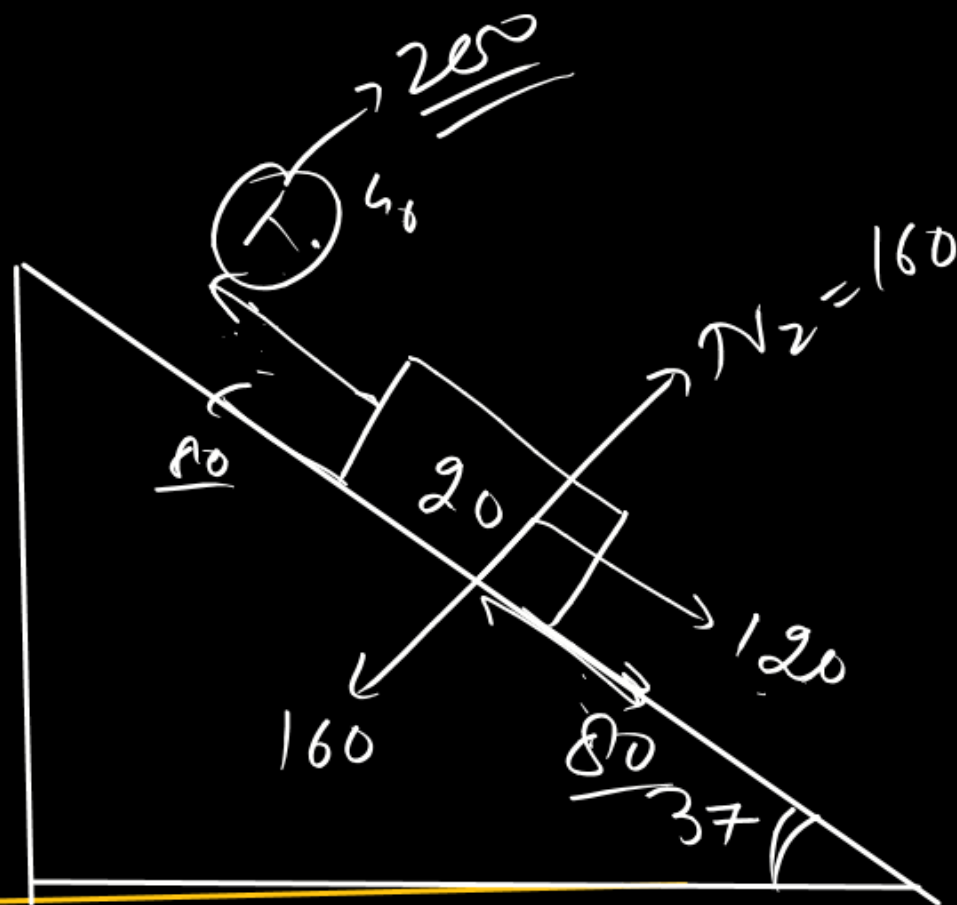
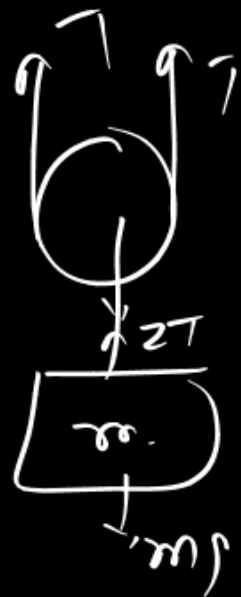
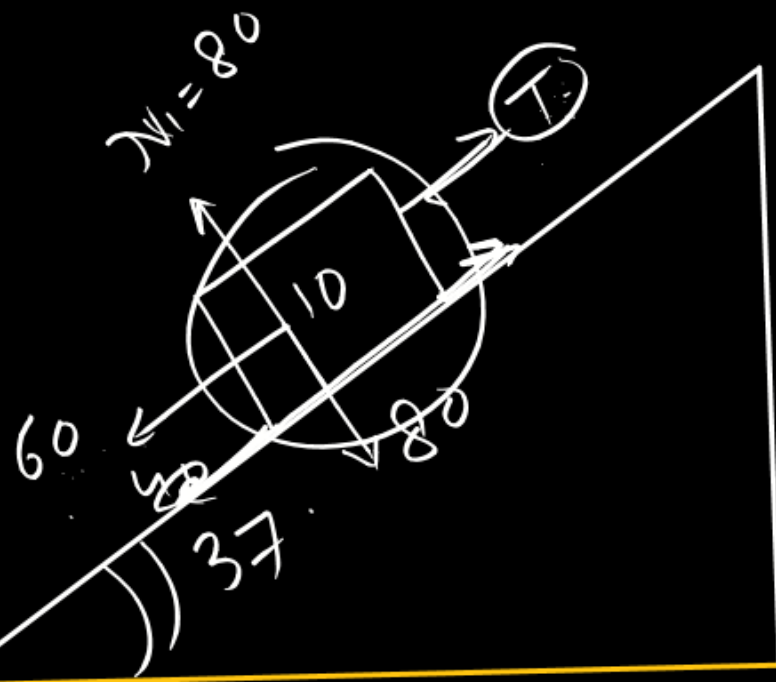
$$T = 40N$$

$$2 \times 40 = m \times 10$$

$$m = 8$$

$$T = 100$$

$$T = 100$$



$$f_{s \max} = 0.5 \times 80$$

$$= 40$$

$$T = 20N$$

$$f_{2 \max} = 0.5 \times 160 = 80$$

$$2 \times 100 = m \times 10 \Rightarrow m = 20$$

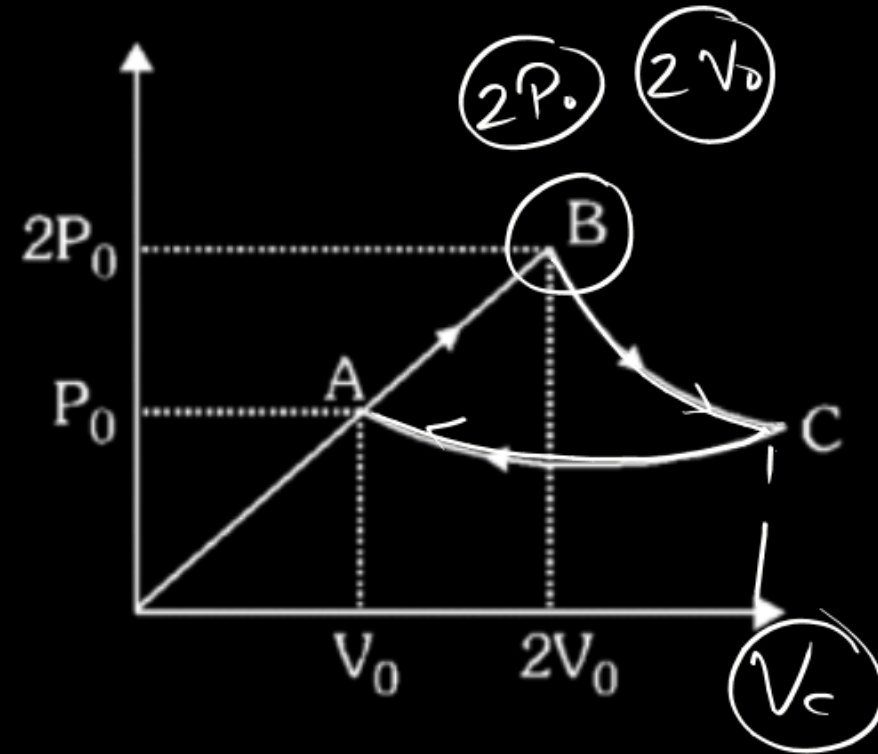
$$20 - 8 = 12$$

$$T = 40$$

One mole of a diatomic gas is taken around a cyclic process shown in the figure. The process BC is adiabatic expansion and the process CA is isothermal compression. If the volume of the gas at C is given by  $V_C = (2)^n V_0$ , then the value of n is

$$\rightarrow f = 5$$

$$\gamma = 1 + \frac{2}{5} = \frac{7}{5}$$



$$\gamma = \frac{\gamma + 1}{\gamma - 1}$$

$$= \frac{\frac{7}{5} + 1}{\frac{7}{5} - 1} = \frac{12}{2} = 6$$

$$(2P_0)(2V_0)^\gamma = P_C V_C^\gamma \quad - (1)$$

$$P_0 V_0 = P_C V_C \quad - (2)$$

$$2^{\gamma+1} V_0^{\gamma+1} = V_C^{\gamma+1}$$

$$V_C = 2^{\frac{\gamma+1}{\gamma-1}} V_0$$

$$= 2^n V_0$$

A particle is thrown with initial speed  $u = 1\text{m/s}$  in vertically upward direction. Due to air resistance, retardation produced in the particle is proportional to square of velocity of particle i.e.  $\propto v^2$ , where  $\alpha$  is  $10\text{ m}^{-1}$ . If time taken by particle to reach maximum height is  $\frac{\pi}{x}$  sec then,  $x$  is (Take  $g = 10\text{m/s}^2$ )

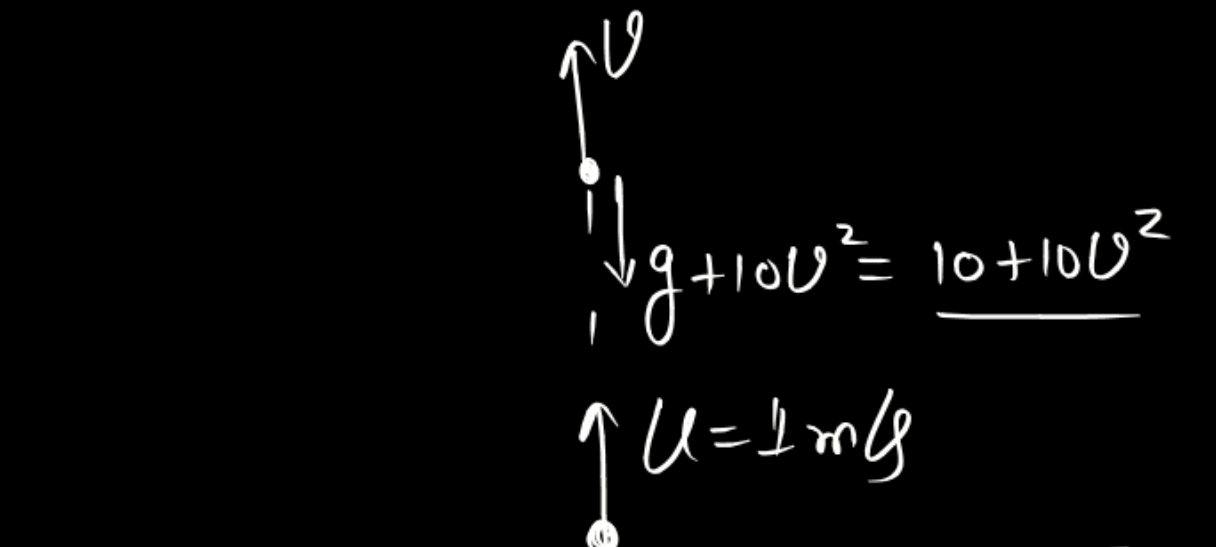


Diagram illustrating the forces acting on the particle during its upward motion. The initial velocity is  $u = 1\text{m/s}$  (upward). The forces acting on the particle are gravity ( $g$ ) and air resistance ( $10v^2$ ), both acting downward. The net retardation is given as  $g + 10v^2 = 10 + 10v^2$ .

$$10t = \frac{\pi}{4} \Rightarrow t = \frac{\pi}{40} \text{ sec}$$

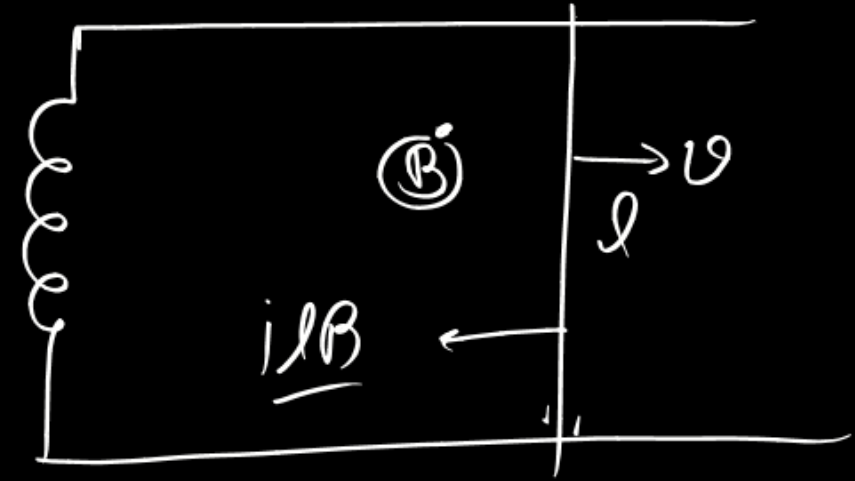
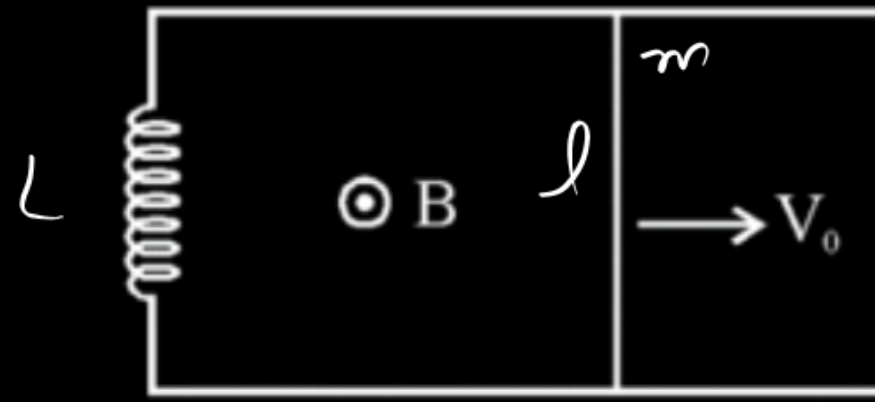
$$= \frac{3.14}{40} \text{ sec}$$

$$-10(1+v^2) = \frac{dv}{dt} = v \frac{dv}{dx}$$

$$-10 \int_0^t dt = \int_1^0 \frac{dv}{1+v^2}$$

$$-10t = \left( \tan^{-1} v \right)_1^0 = \tan^{-1} 0 - \tan^{-1} 1$$

A loop is formed by two parallel conductors connected by a solenoid with inductance  $L = 10^3$  H and a conducting rod of mass  $m = 1$  gm which can freely slide (without friction) over a pair of conductors. The conductors are located in a horizontal plane in a uniform magnetic field  $B = 4$  T in the direction shown. The distance between conductors is equal to  $l = 1$  m. At the moment  $t = 0$ , the rod is imparted an initial velocity  $V_0$  to the right. Determine angular frequency (in rad/s) of oscillation of rod.



$$a = \frac{B^2 l^2}{m L} x \quad \omega^2$$

$$\omega = \sqrt{\frac{B^2 l^2}{m L}}$$

$$= \sqrt{\frac{4^2 \times 1^2}{10^{-3} \times 10^3}}$$

$$= 4 \text{ Rad/sec}$$

$$\mathcal{E} = v B l = L \frac{di}{dt}$$

$$B l \int v dt = L \int di$$

$$B l x = L i$$

$$f = \left( \frac{B l x}{L} \right) l B$$

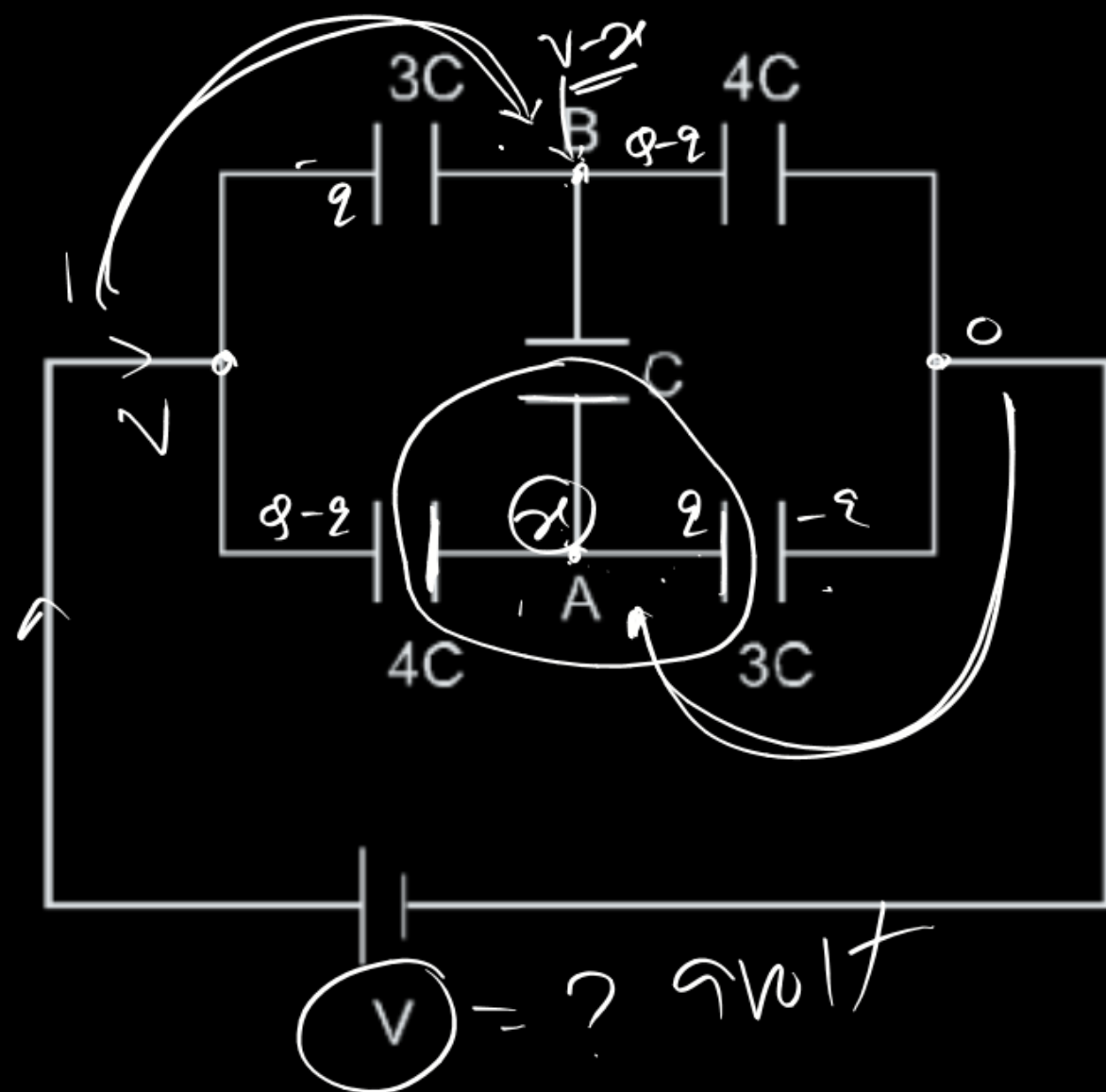
$$f = \frac{B^2 l^2}{L} x \Rightarrow m/s$$

Consider the given circuit. Initially all capacitor were uncharged.

$$4\phi(x-v) + 4\phi(2x-v) + 3\phi(x-0) = 0$$

$$4x - 4v + 2x - v + 3x = 0$$

$$9x = 5V$$



$$(v-2x) - 2x$$

$$\boxed{v - 2x = 1} \text{ given}$$

$$v - 2\left(\frac{5v}{9}\right) = 1$$

$$v - \frac{10v}{9} = 1$$

$$-\frac{v}{9} = 1$$

$$\boxed{v = -9}$$

If  $V_B - V_A = 1$  volt in steady state, the potential difference across the terminals of battery is  $V = 9V$

The variation of density ( $\rho$ ) with temperature ( $T$ ) of an ideal gas of molecular mass ( $M$ ) at constant pressure is plotted as shown in figure. The pressure of the gas at state P is  $\frac{RT^2}{\sqrt{KM}}$ . Find the value of K.

$$PV = \frac{m}{M} RT$$

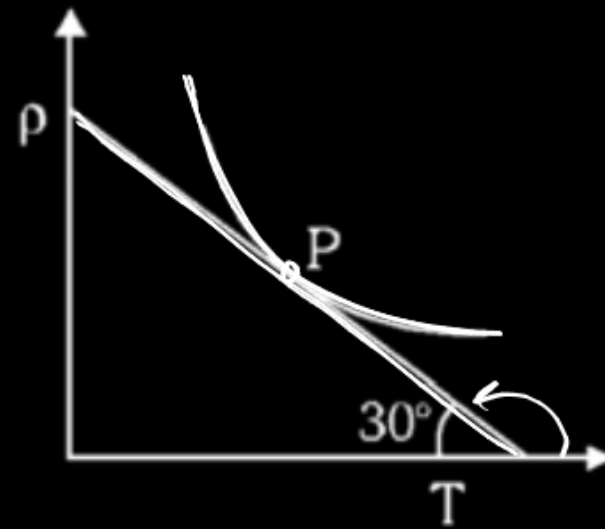
$$PM = \rho RT$$

$$\rho T = \frac{PM}{R} = \text{const}$$

$$\rho T = C$$

$$\rho dT + T d\rho = 0$$

$$\frac{d\rho}{dT} = -\frac{\rho}{T} = \tan 150^\circ = -\frac{1}{\sqrt{3}}$$



$$\rho / T = \frac{d\rho}{dT}$$

$$\frac{\rho}{T} = \frac{1}{\sqrt{3}}$$

$$\begin{aligned} P &= \frac{\rho T R}{M} \\ &= \frac{T^2 R}{M \sqrt{3}} \end{aligned}$$

$$\sqrt{3} = \sqrt{K}$$

$$K = 3$$

In a vernier calliper 10 vernier scale division coincides with 9 main scale division. Where 1 main scale division is 0.1 cm.

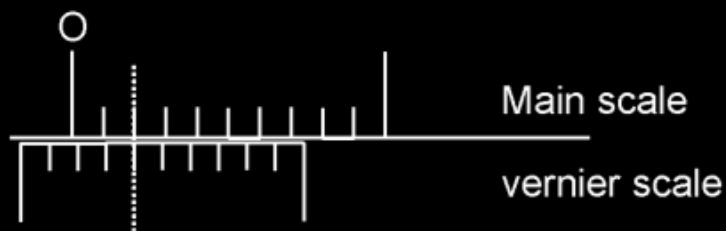


Figure shows main scale and vernier scale when gap between two jaws is zero. The zero error in the faulty vernier calliper is  $x$  (in mm) then,  $5x$  is

Two persons A and B wear glasses of optical powers (in air)  $P_1 = +2\text{ D}$  and  $P_2 = +1\text{ D}$  respectively. The glasses have refractive index 1.5. Now they jump into a swimming pool and look at each other. B appears to be present at distance 2m (from A) to A. A appears to be present at distance 1m (from B) to B. The refractive index of water in the swimming pool, in the form  $\frac{X}{5}$  and find X.